COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 - DATA STRUCTURES AND ALGORITHMS	**SECTION:**	CPE21S1
DATE PERFORMED:	Jul 9, 2026	**DATE SUBMITTED:**	Jul 14, 2026
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ACTIVITY NO. 1.1 [REVIEW OF C++ PROGRAMMING]

1. PROCEDURE OUTPUT

ILO A: Create Code That Follows the Basic C++ Code Structure

For this activity, you have to demonstrate the use of a function prototype. The section on class declaration and method definition will be used for the function prototype and the function will be defined in the follow method definition section after the main function.

A function prototype in c++ is a declaration of the name, parameters and return type of the function before its definition. Write a C++ code the satisfies the following:

Create a function that will take two numbers and display the sum.

Create a function that will return whether variable A is greater than variable B.

Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

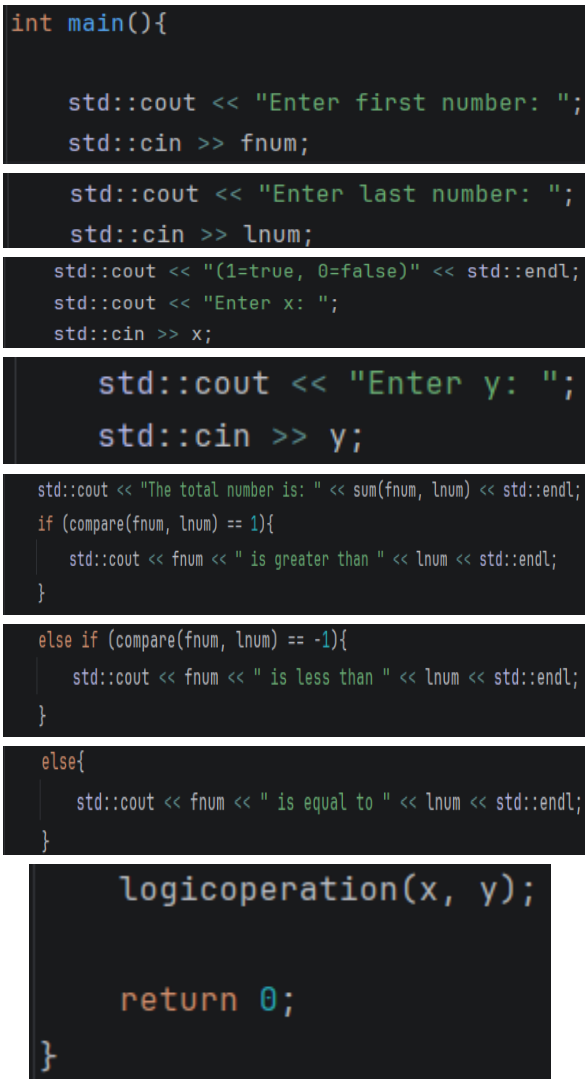
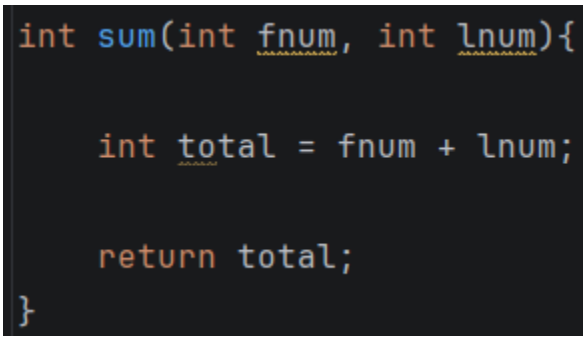
Note:

The driver program must call each function.

The definitions must be after the main function.

SECTION	ANSWER
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`int fnum, lnum;` `bool x, y;`

Class Declaration and Method Definition Section	`int sum(int fnum, int lnum);`
	`int compare(int fnum, int lnum);`
	`bool logicoperation(bool x, bool y);`

Main Function	
Method Definition	

```
int compare(int fnum, int lnum){

    if (fnum > lnum){
        return 1;
    }
    else if (fnum < lnum){
        return -1;
    }
    else{
        return 0;
    }
}
```

```
bool logicoperation(bool x, bool y){

    std::cout << "x AND y: " << (x && y) << std::endl;
    std::cout << "x OR y: " << (x || y) << std::endl;
    std::cout << "NOT x: " << (!x) << std::endl;
    std::cout << "NOT y: " << (!y) << std::endl;

    return 0;
}
```

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

In this section, the initial implementation for a class triangle will be implemented. The step-by-step procedure is shown below:

Step 1. Include the necessary header files. For this one, we only need **#include <iostream>**

Step 2. Create the triangle class. Assign it with private variables: totalAngle, angleA, angleB, and angleC.

class Triangle{
private:
double totalAngle, angleA, angleB, angleC;

Step 3. We then create public methods. The constructor must allow for creation of the object with 3 initial angles to be stored in our previously defined variables angleA, angleB and angleC. Another method has to be made if the user wants to change the initial values, this will also accept 3 arguments to change the values in angleA, angleB and angleC. Lastly, a function to validate whether the given values make our shape an actual triangle.

public:
Triangle(double A, double B, double C);
void setAngles(double A, double B, double C);
const bool validateTriangle();
};

Step 4. Define the methods.

Triangle::Triangle(double A, double B, double C) {

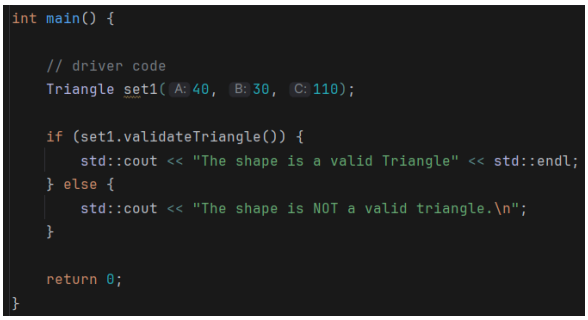
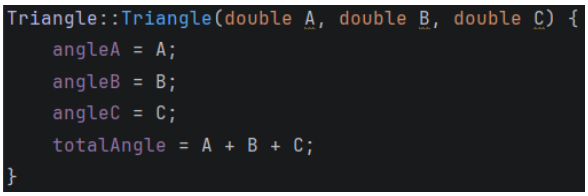
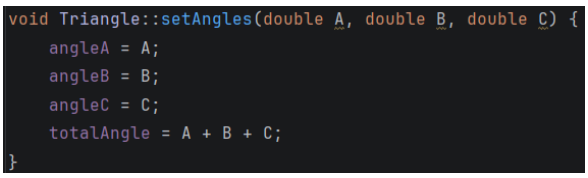
```
angleA = A;
7
angleB = B;
angleC = C;
totalAngle = A+B+C;
}
void Triangle::setAngles(double A, double B, double C) {
angleA = A;
angleB = B;
angleC = C;
totalAngle = A+B+C;
}
const bool Triangle::validateTriangle() {
return (totalAngle <= 180);
}
```

Step 5. Create the driver code.

```
int main(){
//driver code
Triangle set1(40, 30, 110);
if(set1.validateTriangle()){
std::cout << "The shape is a valid triangle.\n";
} else {
std::cout << "The shape is NOT a valid triangle.\n";
}
return 0;
}
```

SECTION	ANSWER
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	none

Class Declaration and Method Definition Section	`class Triangle {`
	`private:`
	`double totalAngle, angleA, angleB, angleC;`
	
	`public:`
	`// constructor`
	`Triangle(double A, double B, double C);`
	
	`// methods`
	`void setAngles(double A, double B, double C);`
	
	`// validation`
	`const bool validateTriangle();`
	`};`

Main Function	
Method Definition	  

Based on the program output, it helps me to understand how the class works. The program can determine the entered angles to satisfy the condition for a triangle. In the program, there are 40, 30, 100 and it is equal to 180 and it is a valid triangle.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C: Solve Different Problems using the C++ Programming Language

The supplementary activities are meant to gauge your ability in using C++. The problems below range from easy to intermediate to advanced problems. Note your difficulties after answering the problems below.

1. Create a C++ program to swap the two numbers in different variables.

```cpp
//Create a C++ program to swap the two numbers in different variables.
#include <iostream>

int main(){

    int fnum,lnum,temp;

    std::cout << "Enter first number: ";
    std::cin >> fnum;

    std::cout << "Enter second number: ";
    std::cin >> lnum;

    temp = fnum;
    fnum = lnum;
    lnum = temp;

    std::cout << "\nAfter swapping the first number and second number, the result is: \n";
    std::cout << "First number = " << fnum << "\nSecond number = " << lnum << std::endl;

    return 0;
}
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

```cpp
//Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.
#include <iostream>

double converter(double kelvin);
double k, f;

double converter(double kelvin){

    double f = (kelvin - 273.15) * 9 / 5 + 32;

    return f;
}
```

```cpp
int main(){

    std::cout << "Enter temperature in Kelvin: ";
    std::cin >> k;

    f = converter(k);
    std::cout << "Temperature in Fahrenheit: " << f << std::endl;

    return 0;
}
```

3. Create a C++ program that has a function that will calculate the distance between two points.

```cpp
//Create a C++ program that has a function that will calculate the distance between two points
#include <iostream>
#include <cmath>

double a1, b1, a2, b2;
double distance;

int main(){

    std::cout << "Enter coordinates x1: ";
    std::cin >> a1;

    std::cout << "Enter coordinates y1: ";
    std::cin >> b1;

    std::cout << "Enter coordinates x1: ";
    std::cin >> a2;
    std::cout << "Enter coordinates y2: ";
    std::cin >> b2;

    distance = sqrt( X: pow( X: a2 - a1,  Y: 2) + pow( X: b2 - b1,  Y: 2));
    std::cout << "Distance is = " << distance << std::endl;

    return 0;
}
```

4. Modify the code given in ILO B and add the following functions:

a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```
#include <iostream>
#include <cmath>

class Triangle {
private:
    double side1, side2, side3;
    double angle1, angle2, angle3;

public:
    Triangle(double s1, double s2, double s3,
             double a1, double a2, double a3);

    bool isValidTriangle();
    double findArea();
    double findPerimeter();
    void displayTriangleType();
};
```

```cpp
// Constructor
Triangle::Triangle(double s1, double s2, double s3,
                   double a1, double a2, double a3) {
    side1 = s1;
    side2 = s2;
    side3 = s3;

    angle1 = a1;
    angle2 = a2;
    angle3 = a3;
}

// Check if triangle is valid
bool Triangle::isValidTriangle() {
    if ((angle1 + angle2 + angle3) == 180) {
        return true;
    }
    return false;
}
```

```cpp
// Compute perimeter
double Triangle::findPerimeter() {
    return side1 + side2 + side3;
}

// Compute area using Heron's Formula
double Triangle::findArea() {
    double semiPerimeter = findPerimeter() / 2.0;

    return std::sqrt( X: semiPerimeter *
                      (semiPerimeter - side1) *
                      (semiPerimeter - side2) *
                      (semiPerimeter - side3));
}
```

```cpp
// Determine triangle type
void Triangle::displayTriangleType() {

    if (angle1 < 90 && angle2 < 90 && angle3 < 90) {
        std::cout << "Triangle Type: Acute Triangle" << std::endl;
    }
    else if (angle1 > 90 || angle2 > 90 || angle3 > 90) {
        std::cout << "Triangle Type: Obtuse Triangle" << std::endl;
    }
    else {
        std::cout << "Triangle Type: Others" << std::endl;
    }
}
```

```
int main() {

    double s1, s2, s3;
    double a1, a2, a3;
    int choice;

    std::cout << "Enter Side 1: ";
    std::cin >> s1;

    std::cout << "Enter Side 2: ";
    std::cin >> s2;

    std::cout << "Enter Side 3: ";
    std::cin >> s3;

    std::cout << "Enter Angle 1: ";
    std::cin >> a1;
```

```
    std::cout << "Enter Angle 2: ";
    std::cin >> a2;

    std::cout << "Enter Angle 3: ";
    std::cin >> a3;

    Triangle t(s1, s2, s3, a1, a2, a3);

    if (!t.isValidTriangle()) {
        std::cout << "\nThe given values do not form a valid triangle." << std::endl;
        return 0;
    }
```

```
    std::cout << "\n===== MENU =====" << std::endl;
    std::cout << "1. Compute Area" << std::endl;
    std::cout << "2. Compute Perimeter" << std::endl;
    std::cout << "3. Determine Triangle Type" << std::endl;
    std::cout << "Enter your choice: ";
    std::cin >> choice;

    switch (choice) {
        case 1:
            std::cout << "Area = " << t.findArea() << std::endl;
            break;

        case 2:
            std::cout << "Perimeter = " << t.findPerimeter() << std::endl;
            break;
        case 3:
            t.displayTriangleType();
            break;

        default:
            std::cout << "Invalid menu choice." << std::endl;
    }

    return 0;
}
```

Conclusion: I learned how to use function prototype and make classes, it helps me to make my program organized and make the code easier to manage and run.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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